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Project Title:	Applications of Electric Fields in Microgravity Combustion Science		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Zip Code:	92697-3975	**Congressional District:**	48
Comments:			
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No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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In order to achieve the project objectives, there are two principal thrusts to the research. One focuses on the electrically controlled flame system, where experiments are conducted in an Earth based laboratory enabling a wide range of non-intrusive diagnostics on flames in various g-loading conditions including that equivalent to microgravity. Diagnostics such as schlieren, shadowgraph, deflection imaging, are applied to visualize a burning methane-air flame under the influence of electrically induced microbuoyancy. Numerical simulations predict the forces involved and allow new designs for enhanced flame control.
For the ion wind driven burner and extinguisher systems, it is necessary to understand the limitations of ion winds. For example, it is to be determined whether or not the airflow can be accelerated with the addition of a nozzle and/or if the blowers are staged. From the results, an ionic air swirler is to be designed where a vortex is formed within a tube containing air at STP with adequate strength to hold a flame. The vortical flow can also be applied to a fan-less

Task Description:	electrostatic precipitator capable of enhanced particle removal efficiency by combining electrostatic and centrifugal forces on the particles entrained into the system. This project explores the use of electric fields for a wide variety of space exploration applications, primarily those involving combustion. The specific objectives of the study include: 1) Experimentally determine the extent to which electric fields can simulate a variable gravity condition (e.g., Mars gravity, moon gravity, zero gravity) for flames in an Earth-based laboratory so as to study phenomena that will be important in these space and planetary environments. 2) Examine the fundamentals of ion wind interactions with flames and their effects on flame behavior in order to understand the practicality of controlling flames electrically. 3) Determine active control methods to attain and maintain effective g-loadings on flames by ion wind actuation and quantify the extent of this action; such a method can be demonstrated by removing g-jitter from combustion experiments in parabolic flights. 4) Construct a dual-electrostatic burner (i.e., no moving parts) in which a flame is sustained and oxygenated by an air swirl generated from an ion wind and liquid fuel is delivered via wick or electrostatic atomization. 5) Optimize ion wind generation for space-based combustion applications and investigate concepts for maximizing flow. 6) Use ion winds to enhance the performance of electrostatic precipitators. 7) Evaluate the potential for hand-held electrosprays to extinguish fires in a variety of gravity-levels. Understanding electric field interactions with flames is central to combustion control, a technology that not has application in advanced propulsion and power systems, but also toward fire safety in space and on other planets. Appropriately applied fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our facility provides a unique ability to make simultaneous measurements of temperature, species concentration, flow patterns, and ion currents. One goal central to NASA’s mission is to improve spacecraft and remote habitat fire safety for travel and off-world activities. The ion wind and electrospray work is expected to reveal new and improved ways to prevent, detect, and extinguish fires. In addition, advances in small-scale combustion systems capable of providing compact power can be developed without moving parts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Taking the specific objectives in turn, the following progress has been achieved over the first year of the project -- • Experimentally determine the extent to which electric fields can simulate a variable gravity condition – A new experimental apparatus was constructed that isolates a small capillary flame from random air currents in the laboratory. The experiment allows full optical access to allow schlieren, chemiluminescence, and other spectroscopic measurements. Using methane fuel, one important finding was the natural instability of the balance condition (the situation where the flame buoyancy and the downward ion wind force cancels). The instability is intermittent and does not appear in the temperature profiles, but is clearly visible in schlieren movies. The instability does not occur for even modest effective gravity conditions. In addition, ion current measurements show that for the simple capillary configuration, the point of force balance occurs at a point when the flame’s ion production is saturated. This coincidence is independent of the spacing between the capillary and the ground pland and is independent of the flow rate of methane (in the low flow rate range examined). This result is interesting, because it must represent a feature of the specific geometry we are using. A flat flame would not behave in this manner. In addition to the methane results, the behavior of several different fuels was examined. The response of the flame followed expectations in that the higher the number of carbons in the fuel, the larger the ion wind force (for a given field). This result is consistent with past findings. In addition, however, we found that the electric field had a profound influence of soot formation, particularly for acetylene flames. It was possible in some circumstances to create a spherical shell of soot within the acetylene flame near the balance condition. Further examination of this phenomenon is underway. • Examine the fundamentals of ion wind interactions with flames – We have begun constructing a comprehensive CFD model of the capillary flame in an electric field using FEMLAB. So far, we have simulated the flame as a constant potential surface that is also a source of ions. We then can predict the path of the ions. The simulations show that there is a limited region over which the ions can extend before they are dragged to the ground electrode. This region coincides very closely to the region we have found through experiment to be the location of a shear layer separating an outer buoyant plume and the fairly stagnant inner balance region. Eventually, we expect a dynamic model that will allow us to simulate the instantaneous response of the flow field to a step change in potential. This will then help identify the sources of the system characteristic times. • Determine active control methods – No specific progress on this objective yet, though we have the equipment ready to implement any control scheme developed. • Construct a dual-electrostatic burner – We have examined the performance of wicks, liquid flow dispersion on rough surfaces, and we have separately examined the generation of ion winds (discussed further below). • Optimize ion wind generation – A substantial effort has been made in determining the dominant features of an ion wind generator based on a corona ion source. The basic configuration is a needle to a ring electric field. The highly charged needle produces ions that are swept toward the ring by the electric field. These ions then produce a wind by dragging along the neutral air molecules. We have found several geometrical sensitivities in this system, including the distance of the needle to the ring, the width of the ring, the accuracy with which the needle is centered, and the overall obstructions to the flow. We have also discovered that staging needle-tube ion wind generators (i.e., stacking them in series) does not produce ever-higher flow rates. There is a limiting flowrate, which indicates that the ion wind process creates an aerodynamic loss, much as does a pipe corner or valve. We are in the process of measuring this loss coefficient and confirming that it follows a typical minor-loss fluid model. We have further found that it is possible to increase the flow velocity of the ion wind by stacking generators in series and then putting a nozzle at the exit. This then creates sufficient pressure to drive the acceleration through the nozzle but keeps the flow through the ion wind stages low enough that the losses do not remove the advantages. We have studied from 1-10 stages and will be publishing results of these studies very soon. • Use ion winds to enhance the performance of electrostatic precipitators – We have studied the use of a swirling flow generated by ion winds for improving the performance of electrostatic precipitation, particularly in the size range that would fall between straight electrostatic precipitators and straight cyclone separators. That is, submicron particles are relatively easy to remove electrostatically, and particles larger than 10 microns are relatively easy to separate inertially. Between these limits, removal of particles is difficult and so we examine a combined electrostatic/cyclonic system based

	on ion winds. The main wind through the precipitator is created by a needle ring ion wind generator. The ions both create the wind and attach to particles. The second, or swirling stage, of the precipitator consists of a cylindrical core electrode with a series of needles jutting out. These needles also produce ions that attach to particles and create a swirling ion wind (the centrifugal component). Talc powder, a controllable and safe source of fine particles (5-100μm), was used to replicate particles found in dusty rooms. It was entrained into the air stream of the precipitator by placing powder onto a vibrating tray placed just below the entrance of the precipitator. The precipitator was run until talc powder accumulated onto the clear acetate walls of the device and the pattern of accumulation was then analyzed qualitatively. The areas very near to the earth rods are lacking in medium to small particles when compared to the areas away from the rods. Otherwise, the distribution of medium and small particles is fairly even over the acetate sheet. There is a slight gradation when moving towards the top of the precipitator, where fewer small to medium particles are found. However, this gradation is more exaggerated for the large particles, so it is difficult to spot large particles higher up the precipitator. It was possible to run the precipitator so that no visible particles escaped the device. As expected, it was necessary to raise the electric field in both stages together to prevent particles from escaping. This is because the higher the linear wind flow through the system (i.e., the shorter the residence time), the stronger the centrifugal component needs to be. The amount of talc lost from the powder tray was compared to the amount collected by the precipitator. The efficiency of the precipitator was effectively 100%. It remains to be determined if this performance is more effective than standard blower designs, but the fact that it has no moving parts and that it is all electric, makes it an attractive possibility for space application. • Evaluate the potential for hand-held electrosprays to extinguish fires – We have not yet examined this feature of the project. We are, however, looking at electrosprays of biologically active materials to see if they remain viable after processing. This is a cross-over application the application of thin films for topical medical purposes can be accomplished.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
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